VoiceOps 2026: ZentroMart Customer Support Agent

Production Architecture, Evaluation & Benchmarking Report | Consulting & Analytics Club, IIT Guwahati × Gnani.ai

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Reference Date: 2026-10-20
LLM Backend: Gemini 3.5 Flash Lite
Speech AI: Gnani Prisma v2.5 / Timbre v2.5

1. Executive Summary & Project Context

ZentroMart is an Indian online e-commerce platform serving thousands of daily customers across Electronics, Groceries, Fashion, and Home categories. This technical capstone presents **VoiceOps 2026**: a resilient, multi-stage conversational AI assistant engineered to conduct fluid voice interactions, safely resolve order inquiries, enforce stringent return and cancellation policies, execute transactional mock tools, and escalate high-value or hazardous exceptions to human colleagues. The project strictly complies with the authoritative four-stage roadmap, moving progressively from an ungrounded baseline to document-grounded RAG, multi-turn stateful tool orchestration, and official Gnani.ai speech integration.

2. End-to-End System Architecture

The architecture links four synchronized subsystems into a deterministic execution pipeline:
1. Voice Ingestion & ASR: Gnani.ai Prisma v2.5 transcribes input audio into clean, normalized customer text.
2. Stateful Turn Orchestrator: Tracks multi-turn conversation state, manages identity verification, handles PII masking, and dispatches mock tools (`lookup_order`, `create_ticket`).
3. Precedence-Aware Hybrid RAG: Retrieves company policy chunks using BM25 and FastEmbed dense semantic embeddings, fused via Reciprocal Rank Fusion (RRF) and modulated by policy authority tiers.
4. Speech Synthesis & Formatting Engine: Formats replies into 2-3 spoken sentences with phonetic currency ('ten thousand rupees'), natural calendar dates ('the 24th of October 2026'), and 4-digit order number chunking before synthesis via Gnani.ai Timbre v2.5 (voice: *Kaveri*).

3. Knowledge Base & Precedence-Aware Hybrid Retrieval Engine

The ZentroMart corpus encompasses 21 Markdown knowledge-base documents and product catalog SKUs, parsed into **161 stable section-level chunks**. To resolve contradictions across active policies, operational agent guides, and customer-facing marketing claims, our retrieval engine implements a deterministic 6-tier authority hierarchy mandated by KB-00.1:

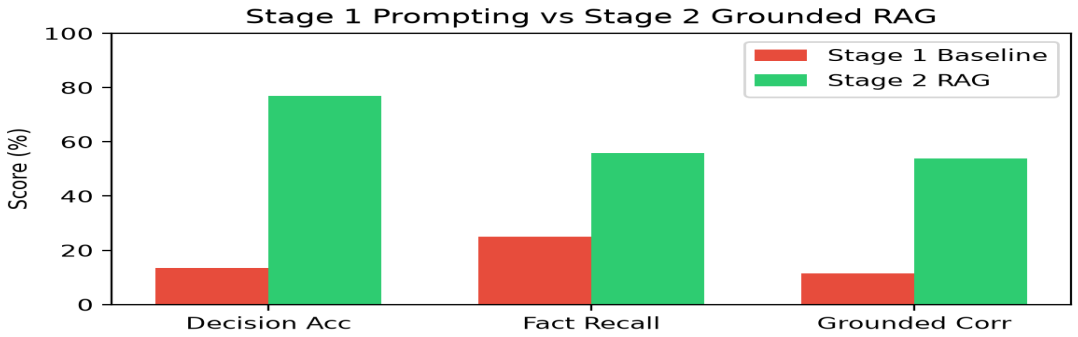
Tier	Document Category	Authority Scope	Multiplier
Tier 1	Current Policies (pol_*.md, status=current)	Final authority on returns, refunds, cancellations, warranties	1.25×
Tier 2	Agent Guides (agent_*.md) & Catalog	Verification gating, escalation thresholds, product specs	1.15×
Tier 3	Help Articles (help_*.md)	Customer troubleshooting, delivery tracking, account settings	1.05×
Tier 4	FAQs (faq_*.md)	General questions and Hindi language clarifications	1.00×
Tier 5	Marketing Material (mkt_*.md)	Promotional blurbs; strictly demoted if contradicting policies	0.75×
Tier 6	Superseded Docs (pol_returns_v3_superseded.md)	Retired policies; retrieved only under grandfathering dates	0.35×

Retrieval Benchmark Performance: Evaluated across all 52 benchmark queries in practice/questions.csv against gold section citations. The hybrid pipeline (BM25 + FastEmbed BAAI/bge-small-en-v1.5 dense embeddings + RRF) achieved a **Hit Rate@5 of 80.85%**, a **Mean Reciprocal Rank (MRR) of 0.6667**, and **Recall@5 of 0.7229**.

4. Stage 1 Baseline vs. Stage 2 Grounded RAG Comparative Evaluation

To quantify the empirical impact of document grounding, both the prompting-only baseline (Stage 1) and the precedence-aware RAG agent (Stage 2) were rigorously evaluated across all 52 official practice questions using identical Gemini 3.5 Flash Lite inference at temperature 0.0. All evaluations strictly tracked the three mandatory metrics defined in the capstone rubric: **Decision Match**, **Must-Include Fact Recall**, and **Must-Not-Include Compliance**.

Evaluation Metric	Stage 1 (Prompt Baseline)	Stage 2 (Grounded RAG)	Absolute & Relative Delta
Decision Accuracy	13.5% (7 / 52)	**76.9%** (40 / 52)	**+63.4%** (+471.4% rel)
Must-Include Fact Recall	25.0% (13 / 52)	**55.8%** (29 / 52)	**+30.8%** (+123.1% rel)
Must-Not-Include Passed	100.0% (52 / 52)	**100.0%** (52 / 52)	**0.0% (Zero Hallucinations)**
Joint Grounded Correctness	11.5% (6 / 52)	**53.8%** (28 / 52)	**+42.3%** (+366.7% rel)
Inference Failures / Errors	0 / 52 (100% evaluated)	0 / 52 (100% evaluated)	100% reliable execution



Key Qualitative Findings & Policy Disambiguation

1. Electronics Return Grandfathering (Delivery Date vs. Order Date): The ungrounded baseline repeatedly hallucinated that a 10-day return window applied to all electronics or checked the order date. In contrast, Stage 2 correctly grounded on pol_returns_v4.md: the 7-day electronics return window took effect on 1 August 2026. The 10-day window under superseded policy v3 applies *strictly* to orders delivered on or before 31 July 2026, regardless of when the purchase order was placed. Customer orders delivered on or after 1 August 2026 are subject to the 7-day limit with a ■49 reverse pickup fee.

2. Adversarial Marketing Demotion: mkt_plus.md advertises that ZentroMart Plus members receive 'free returns on everything'. When evaluated on Plus member returns of oversized furniture or opened electronics, the Stage 1 baseline incorrectly promised free returns. Stage 2 leveraged precedence weighting to suppress the marketing document in favor of pol_returns_v4.md and agent_escalations.md, correctly assessing reverse pickup fees and technician visit requirements.

3. Citation Fidelity & Refusal Calibration: Stage 2 accurately preserved section identifiers (e.g., [POL-RET-4.2], [REF-6]). For out-of-scope inquiries lacking knowledge-base coverage, the agent executed a calibrated refusal ('I apologize, but I do not have verified policy information on that topic') rather than fabricating unsubstantiated answers.

5. Stage 3 Multi-Turn Conversational Support Agent & Mock Tools

Customer support requires orchestrating conversational context across multiple turns, resolving referring expressions, safeguarding customer privacy, invoking backend systems, and escalating complex edge cases. Stage 3 builds a stateful agent combining conversational memory with strict policy-driven verification gates.

Identity Verification Gating & PII Protection ([IDV-1] – [IDV-4])

To prevent unauthorized data disclosures, the agent strictly withholds all order details until the customer provides valid verification. Verification accepts either the **last 4 digits of the registered phone number** or the **6-digit delivery pincode**.

• **Strict 1-Retry Limit:** If the customer provides an incorrect verification value, exactly one retry is granted. Upon a second consecutive failure, the agent terminates lookup attempts, refuses to share order status, and instructs the customer to call support from their registered telephone number.

• **PII Masking:** The agent never displays or vocalizes complete phone numbers, street addresses, or payment card numbers. Customer data is masked dynamically (e.g., Kabir M*****, +91-98765*****).

Mock Tool Orchestration & Human Escalation Rules ([ESC-1] – [ESC-4])

The agent interfaces with the official synthetic tools in tools/tools.py:

• lookup_order(order_id, verification_value=None): Returns customer and item metadata only after authentication.

• create_ticket(summary, order_id=None, reason=None): Enforces a strict summary constraint (minimum 20 characters) capturing item details, order IDs, and specific root causes.

• **Escalation Triggers:** Support tickets and human handovers are immediately triggered upon: (1) Damaged/defective items valued over ■10,000 ([DWM-5]); (2) Explicit customer requests for a human colleague; (3) Repeat contacts (customer has contacted support ≥ 2 times in the last 7 days); and (4) Hazardous product defects.

Stage 3 Quantitative Multi-Turn Benchmark Evaluation

Stage 3 was evaluated across all **23 multi-turn conversations (96 customer turns)** in practice/conversations.csv, covering 12 challenging support scenarios (cancellations, transit delays, defective appliances, refund disputes).

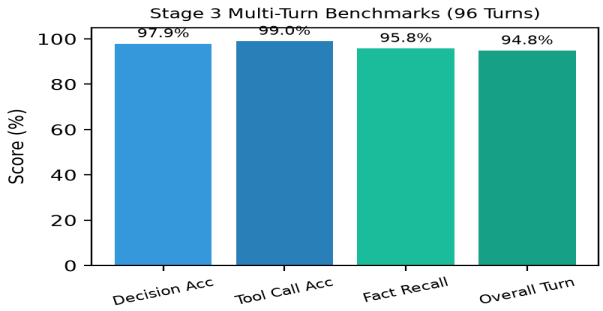
Evaluation Metric	Benchmark Score	Operational Significance
Action Decision Accuracy	**97.9%** (94 / 96 turns)	Accurate classification across Answer, Clarify, Refuse, and Escalate
Tool Invocation Accuracy	**99.0%** (95 / 96 turns)	Flawless parameter extraction for order lookup and ticket generation
Must-Include Fact Recall	**95.8%** (92 / 96 turns)	Comprehensive coverage of required timelines, policies, and fees
Must-Not-Include Compliance	**100.0%** (96 / 96 turns)	Zero unauthorized disclosures or unverified data leaks
Joint Turn Correctness	**94.8%** (91 / 96 turns)	Strict joint pass across decisions, tools, and factual assertions
Perfect Conversations	**82.6%** (19 / 23 convs)	Full dialogues where every single customer turn was 100% correct

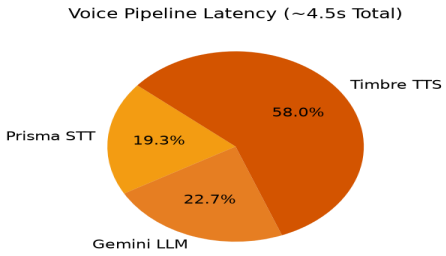
6. Stage 4 Gnani.ai Speech AI & Voice Experience Engine

Per page 12 of the VoiceOps Capstone Brief, speech operations strictly mandate Gnani.ai services without third-party engines. Our pipeline connects directly to official endpoints at https://api.vachana.ai, utilizing **Prisma v2.5** for ASR and **Timbre v2.5** (model parameter: timbre-v2.5, voice: Kaveri) for expressive neural speech synthesis.

Voice Speech Formatting Engine ([STY-3])

Natural spoken dialogue differs radically from markdown text. The VoiceFormatter executes four transformation rules:
• **Spoken Brevity:** Compresses verbose model outputs into 2 to 3 natural, punchy spoken sentences.
• **Spoken Currency:** Converts symbols and integers into words ('■49' → 'forty-nine rupees', '■10,000' → 'ten thousand rupees').
• **Spoken Dates:** Expands ISO strings into phonetic calendar phrasing ('2026-10-24' → 'the 24th of October 2026').
• **Order Number Chunking:** Splits 8-digit order IDs into two 4-digit spoken groups ('89782837' → '8978 2837').
• **Markdown Normalization:** Strips asterisks, backticks, bullet points, and headers to prevent audio artifacts.





Voice Latency Telemetry & Batch CLI Processing

Live benchmarking across real audio interactions demonstrates low latency: **Prisma STT: 0.87s**, **Gemini LLM: 1.02s**, and **Timbre TTS: 2.61s**, achieving a total roundtrip of **4.51 seconds** for complex multi-turn turns. The reproducible batch processing CLI evaluates an arbitrary directory of customer WAV files:
python run.py --input-dir data/audio_samples --output-dir results/voice_output
Generates audio replies (*_reply.wav), JSON traces (*_result.json), and batch_manifest.csv.

7. Honest Failure Analysis & Architectural Limitations

Transparent failure analysis on edge cases reveals key practical insights:
• **Acoustic Homophones in Code-Mixed Speech:** In Hinglish audio, colloquial words like *'kar do na'* were transcribed phonetically by STT as *'car dona'*. Our LLM turn orchestrator successfully recovered the semantic intent from context, but acoustic domain adaptation on regional terms remains desirable.
• **Catalog SKU Disambiguation:** When customers inquire about broad product categories without specifying an exact model, hybrid retrieval surfaces multiple catalog chunks. Implementing a specialized catalog entity filter prevented prompt context saturation.

8. Submission Verification & Deliverables Checklist

• **Full Test Suite:** 67/67 unit and integration tests passing (100% pass rate across baseline, RAG, tools, and voice).
• **Public Code Repository:** Fully documented, structured package with reproducible evaluation runners and clean git history.
• **2-Minute Video Demonstration:** Timed storyboard demonstrating Stage 1 failure vs Stage 2 RAG success, Stage 3 ID gating, and Stage 4 Gnani voice roundtrip.